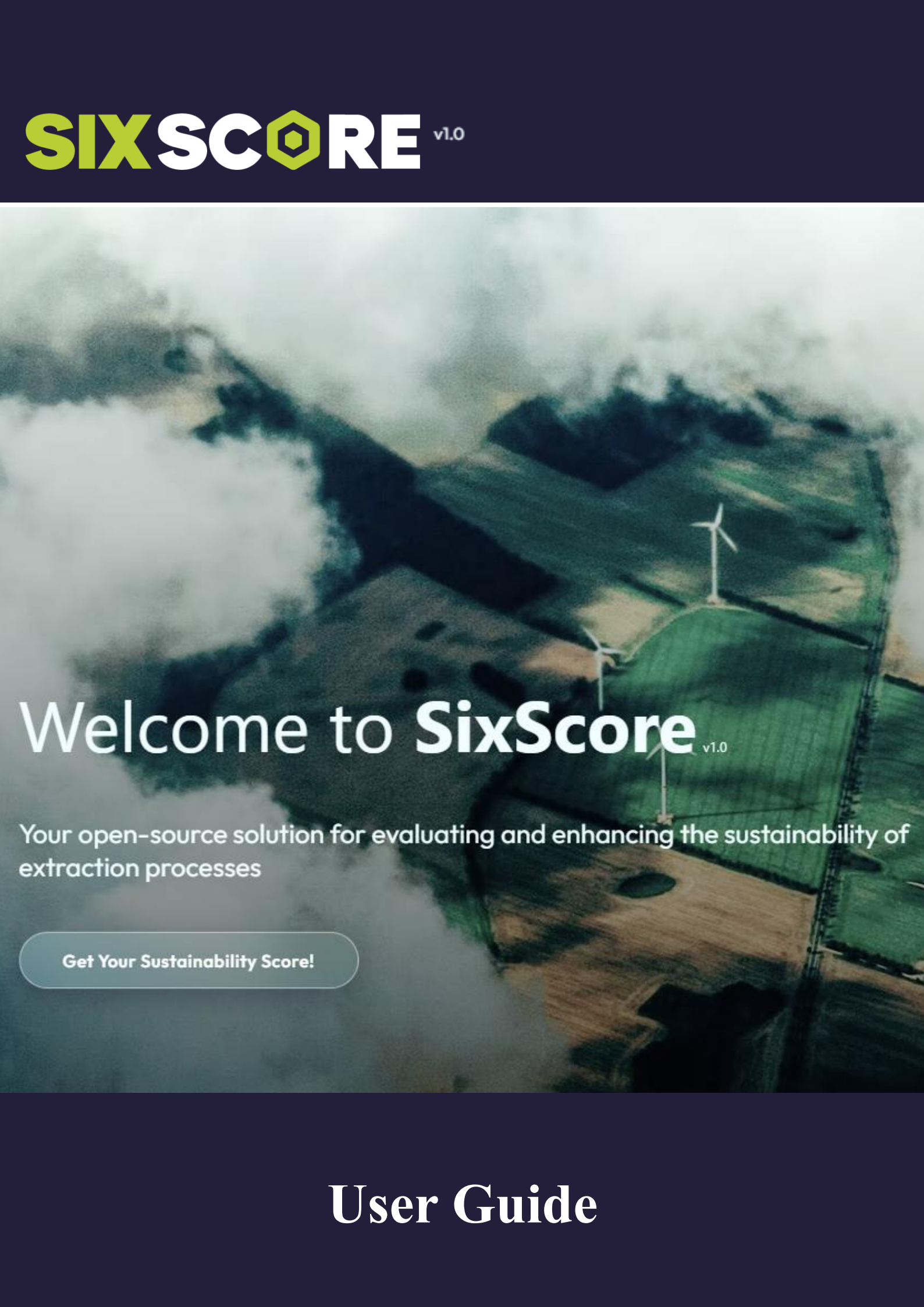




SIXSCORE v1.0

Welcome to SixScore v1.0

Your open-source solution for evaluating and enhancing the sustainability of extraction processes

[Get Your Sustainability Score!](#)

User Guide

SIX Score Hub

Technical Guide for Sustainable Extraction Evaluation

Welcome to the User Guide for **SIX Score**.

This guide is designed to support you in the correct use of the *Sustainable Impact for eXtraction Score* (*SIX Score*), a quantitative tool developed to evaluate and compare the sustainability performance of extraction processes.

The SIX Score integrates six Elemental Factors (EFs) aligned with the principles of Green Extraction as defined by Chemat *et al.*, (doi: 10.3390/ijms13078615) enabling users to evaluate process efficiency, environmental performance, and extract quality within a single, comprehensive framework.

* In this guide, the term ‘*User*’ refers to any individual using the SIX Score software to perform sustainability assessments of extraction processes. The term ‘*Authors*’ refers to the developers of the SIX Score software who designed the methodology and the benchmark system.

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1. Access to the Evaluation Tool

The SIX Score evaluation tool is available online at the following link:

[SIX Score Hub - Homepage](#)

Within the compilation form, several information points are available next to specific fields. These are marked with an information icon “” and provide guidance on how to correctly enter the required data. This guide offers additional explanations to ensure clarity and provide further instruction in case of uncertainty.

2. Required Experimental Data

Before starting the evaluation, ensure that you have the following experimental data available. The additional inputs required are dependent on your experimental outcomes.

- Total amount of target compounds presents in the raw matrix (e.g., polyphenols per unit of biomass)
- Extraction yield of the target compounds (e.g., extracted polyphenols per unit of biomass)
- Total extract mass obtained per unit of matrix
- Organic content percentage in both the biomass and the final extract (See Section 4.3)
- Energy consumption (or instrument energy label) during the extraction step
- Extract selectivity toward the target compounds (e.g., extracted polyphenols per unit of extract)

3. Homepage

The website opens with a homepage that provides an overview of the SIX Score framework (**Figure 1**). By scrolling through the page, you can explore the objectives of the evaluation, understand how the methodology is structured, and review the key features introduced within the tool.

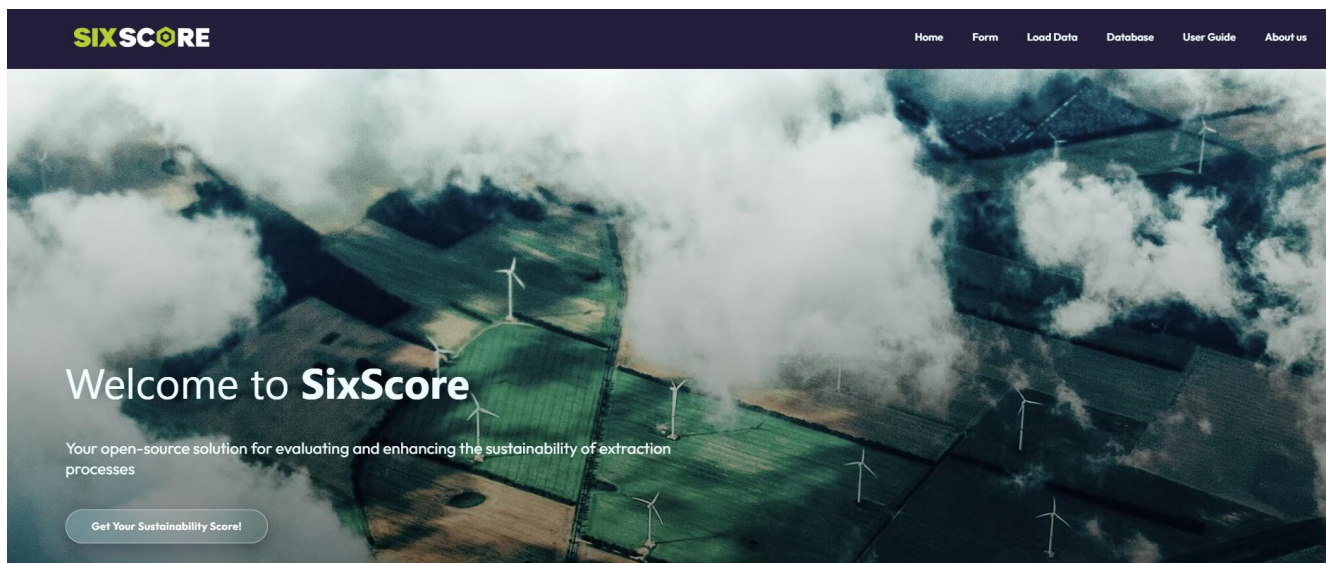


Figure 1. Six Score Homepage.

In addition to the *Home* section, five main menu items are available:

- *Form* – Access the sustainability evaluation form.
- *Load Data* – Click to upload and open a previously completed evaluation saved as a .json file.
- *Database* – View a list of analyses performed by the global SIX Score community and shared voluntarily by Users.
- *User Guide* – Download and consult this guide.
- *About Us* – Learn more about the SIX Score team and the project background.

To begin the sustainability evaluation, click on **Form**.

4. Evaluation Form for SIX Score Assessment

On this page, users can initiate the sustainability assessment of their designed extraction process. The first step is to complete the identification details for the extraction process you are willing to evaluate (**Figure 2**). This information will be used to generate the name of the downloadable results file and, if you choose to share your analysis, to classify and archive it within the SIX Score database.

If you decide to share your data with the community, you will be required to provide the DOI of your publication. This ensures a dual benefit: only peer-reviewed processes are included in the dataset, and interested Users can directly access your article to explore your work in greater detail. For more details about the Database section, see Chapter 7.

Institution Name:

Matrix Name:

Protocol Title:

DOI

Needed for database submission.

Figure 2. List of identification details required for the extraction process (DOI is optional).

4.1 Raw Material

This section concerns the definition of the extraction matrix. Three parameters must be completed: **Feedstock Generation**, **Availability** and **Seasonality** (**Figure 3**). The maximum attributable value for this EF is 5, distributed as follows: up to 2 points for feedstock generation, up to 2 points for availability, and up to 1 point for seasonality. The total collected value is then multiplied by 20 and subtracted from 100 to obtain the final score for this EF.

Raw Material

Feedstock Generation: ⓘ

Select.

Availability: ⓘ

Select.

Seasonability: ⓘ

Select.

Figure 3. Compilation window for the categories used to evaluate the EF Raw Material.

Feedstock Generation:

Following the biorefinery classification concept, the User must identify the type of biomass according to one of the categories listed below (**Table 1**).

Table 1. Category, Description and Attributed Value for Feedstock Generation subcategory.

Category		Description	Attributed Value
1st Generation	Primary crops intended for food or feed consumption		2
2nd Generation	Agricultural or industrial residues		0
3rd Generation	Algae and microorganisms		1
4th Generation	Engineered algae, cyanobacteria, or crops		1

First-generation feedstocks receive the highest penalty due to competition with the food and feed supply chain. Second-generation feedstocks (residues and by-products) are not penalized, as their valorisation aligns strongly with circular economy principles.

Availability:

It is assessed based on the global annual production of the primary crop from which the biomass is derived. To determine whether the biomass falls within the high- or low-availability benchmark, the User should verify the global production data of the related crop.

For example, when valorising grape pomace, grape pruning, or grape seeds, the User must refer to global grape production data. A threshold of 10 million tons per year has been established: feedstocks derived from crops below this value are classified as low availability and penalized, while those above it are considered high availability and receive no penalty.

Seasonality:

Seasonality accounts for limitations related to the availability window of the biomass source. Species with an availability period limited to one season per year or less are penalized, as this may restrict process continuity and industrial scalability.

For example, pomegranate, typically available only in October, would receive a penalty (attributed value = 1, corresponding to 20 penalty points). In contrast, potato skins, derived from a crop with a broader availability and processing window, do not incur penalty points for seasonality.

If the biomass is produced in controlled, non-natural systems such as closed greenhouses, bioreactors, or similar regulated cultivation environments, it is considered continuously available and therefore not subject to seasonality penalties. This includes, for example, *Spirulina* algae cultivated in photobioreactors, soilless or hydroponic specialty crops grown under controlled conditions, and engineered microorganisms or bacteria produced in fermentation systems. In the Supporting Information of the SIX Score manuscript, it is possible to find bibliographic values related to the threshold definition.

4.2 Solvents & Additives

In this section, the User is required to enter all chemicals employed during the process, including those used in pre-treatment, extraction, and purification stages.

Chemicals are divided into two categories: solvents and additives (**Figure 4**). This distinction is necessary because their quantities are evaluated using different calculation scales. Despite this separation, the hazardousness assessment follows the same criteria for both categories.

Hazard classification is determined based on the GHS pictograms, the signal word, and occupational exposure considerations related to vapour or gas emissions.

Solvents & Additives ☑

Solvent List

Add item +

Additive List

Add item +

Figure 4. Solvents & Additives.

Each chemical must be entered in the designated input field. Once inserted, a dedicated compilation block automatically appears for that specific substance (**Figure 5**).

Ethanol

Remove

 Exclamation mark

☒

 Flame

☒

 Environment

☐

 Corrosion

☐

 Flame over circle

☐

 Skull & crossbones

☐

 Health hazard

☐

 Exploding bomb

☐

Signal word

☐ None ☐ Warning ☒ Danger

Vapour/gas emission

☒

Amount (S:L)

1:

Citric Acid

Remove

 Exclamation mark

☒

 Flame

☐

 Environment

☐

 Corrosion

☐

 Flame over circle

☐

 Skull & crossbones

☐

 Health hazard

☐

 Exploding bomb

☐

Signal word

☐ None ☒ Warning ☐ Danger

Vapour/gas emission

☐

Amount (% on biomass weight)

Figure 5. Dedicated compilation blocks for Solvents & Additives.

When the hazardousness assessment screen appears, the User must assign the relevant hazard points by selecting the appropriate GHS pictograms, signal word, and vapour or gas emission classification. Information on the GHS classification and signal word can be obtained from the chemical's Safety

Data Sheet (SDS). The inclusion of vapour or gas emission is determined using a vapour pressure threshold of 0.5 kPa at 20 °C, as established by the REACH regulation.

Finally, the quantity of each chemical must be entered relative to the processed biomass. For solvents, the amount should be expressed as a mass-to-volume ratio with respect to the biomass. For additives, the quantity must be expressed as weight percentage relative to the total biomass.

Certain specific scenarios require methodological clarification. Supercritical fluids and deep eutectic solvents represent two exceptions to the standard liquid-based ratio approach, and in both cases, mass must be used instead of volume.

In the case of supercritical fluids, solvent volume varies as a function of applied pressure and is therefore not a stable reference parameter. For consistency and normalization purposes, the mass of the gas should be used. The mass can be calculated using the equation $PV = ZnRT$, where Z is the compressibility factor characteristic of the gas, determined according to the *Peng–Robinson* equation of state.

Deep eutectic solvents must be decomposed into their individual components, and each component should be entered separately in relation to the biomass. Here again, quantities must be expressed on a mass basis rather than as liquid volume.

These two cases are treated as methodological exceptions within the benchmark system. The possibility of expressing all process ratios exclusively on a mass-to-mass basis, including the solvent-to-biomass ratio, was carefully evaluated during the methodological development. However, such an approach was not adopted, as it would introduce systematic bias. Specifically, solvents with lower density (e.g., hydrocarbons or ethers) would appear quantitatively less impactful than denser solvents (e.g., water) when expressed solely on a mass basis, despite not necessarily representing a more sustainable option. In the Supporting Information of the SIX Score manuscript, it is possible to find tabulated values for main solvent entries.

4.3 Extraction

The required fields to be completed in this section are: *Depletion Efficiency*, *Solvent Impact*, *Carbon Economy*, and *Extraction Duration*. Each subcategory requires specific input data, as described below.

Depletion Efficiency:

Depletion Efficiency

Overall target content mg target compound / g matrix ⓘ

Insert a valid numerical value

Target yield mg target compound / g matrix

Insert a valid numerical value

Figure 6. Depletion efficiency compilation window

Depletion efficiency requires knowledge of the total amount of target compounds present in the biomass (**Figure 6**). To determine this value, the User may employ a conventional extraction method of their choice. The authors recommend using methods recognized in standard literature, such as those reported in official Pharmacopoeias, or performing an exhaustive sequential extraction to establish the maximum extractable amount. In general, the chosen method should align with the User's objectives for comparison. This value serves as the reference for calculating the actual depletion achieved by the process under investigation.

Solvent Impact:

Solvent impact

N° of bio-derived solvents

Insert a valid numerical value

N° of petrol-based solvents

Insert a valid numerical value

Figure 7. Solvent Impact compilation window.

The solvent impact parameter requires the User to indicate the total number of solvents employed during the process, distinguishing between bio-derived and petroleum-based solvents (**Figure 7**). The software assigns a penalty proportional to the total number of solvents used, with a higher penalty applied to petroleum-based solvents due to their greater environmental impact. This parameter therefore rewards solvent minimization and the adoption of renewable alternatives.

Carbon Economy:

Carbon economy ⓘ

Product g Org. Fr./g Extract

Insert a valid numerical value

Starting matrix g Org. Fr./g Matrix

Insert a valid numerical value

Total yield g Extract/g Matrix

Insert a valid numerical value

Figure 8. Carbon Economy compilation window

The carbon economy assessment requires the insertion of two values obtained through calcination procedures aimed at distinguishing the organic fraction from the inorganic fraction (**Figure 8**). To determine these values, the authors recommend a standardized protocol consisting of an initial dehydration step at 100 °C for 12 hours to determine the dry weight of the sample, followed by calcination at 650 °C for 4 hours. This procedure allows quantification of the residual inorganic fraction. By subtracting the inorganic residue from the dry weight, the organic fraction can be calculated. The User is then required to enter the overall process yield based on the initial biomass. The software calculates the carbon economy according to the following equation:

$$CE = \frac{\left[\text{Tot Yield} \left(\frac{g \text{ extr.}}{g \text{ matrix}} \right) \cdot \text{Extr.Org.Fract.} \left(\frac{g \text{ org. fract.}}{g \text{ extr.}} \right) \right]}{\text{Matrix Org.Fract.} \left(\frac{g \text{ org. fract.}}{g \text{ matrix}} \right)}$$

This calculation expresses how efficiently the organic carbon present in the biomass is transferred into the final extract.

Extraction Duration:

Time efficiency

Duration minutes

Insert a valid numerical value

Figure 9. Extraction duration compilation window

The extraction duration is expressed in minutes and refers exclusively to the extraction step (**Figure 9**). Pre-treatment operations and downstream processing times must not be included. In the case of continuous flow extraction systems, the time must correspond to the residence time, meaning the effective time normalized to the treated biomass volume during a single passage through the system.

4.4 Process & Equipment

The screenshot displays the 'Processes & Equipment' window. At the top, there is a title bar with a toggle switch on the right. Below the title bar, the word 'Processes' is followed by a dropdown menu currently showing 'Other...' and a text input field labeled 'process name'. To the right of the text field is a green 'Add' button with a plus icon. Below this, the window title 'Processes & Equipment' is repeated with another toggle switch. The main area is titled 'Processes' and contains a large dropdown menu labeled 'Select process'. This menu is open, showing a list of predefined processes: Grinding, Sieving, Extraction, Microwaves, Ultrasound, Hydrodynamic Cavitation, Pulsed Electric Field, Intense Pulse Light, Instant Controlled Pressure Drop, Extrusion, High Shear Homogenization, Tribology, Ohmic Heating, Electric Resistance Heating, De/Pressurization Only, Cooling/Freezing, Ambient Conditions, and Downstream. To the right of the dropdown menu, there are three horizontal bars, each with a toggle switch. The first and third bars have their switches turned on, while the second bar has its switch turned off. A green 'Add' button with a plus icon is located to the right of the dropdown menu.

Figure 10. Process & Equipment selection of process window.

Within the P&E section, Users can select different process from a drop-down menu (**Figure 10**). The listed processes have been predefined by the authors, as they represent the most commonly used techniques in the field of extraction. If your specific process is not included in the list, you may select “Other...” and manually enter the process name. This ensures flexibility while maintaining standardization for the most widely adopted technologies. Once the process is selected and confirmed, the data entry window for the selected technique will appear (**Figure 11**). In this section, the user must provide the relative operating *Pressure Range*, *Energy Consumption* and *Duration* of the equipment.

Ultrasound

Remove 

Pressure range Low: 0  mbar High: 0  bar

Energy consumption kWh/kg 

Duration minutes

Figure 11. Process characteristics compilation window.

Pressure Range:

Pressure values must be reported as read on the pressure gauge. Therefore, a value of 0 corresponds to atmospheric pressure.

Energy Consumption:

Energy consumption must be entered as a value normalized to the quantity of biomass processed. This normalization ensures comparability across systems with different processing capacities and prevents bias against larger-scale equipment. For instance, when multiple instruments of varying sizes are used, *i.e. a 2 kg capacity oven for drying and a 50 g capacity ultrasound device*, normalizing energy per unit of treated biomass allows meaningful comparison of energy efficiency between these different pieces of equipment within the same process. In the Supporting Information of the SIX Score manuscript, it is possible to find tabulated values for main extraction technologies.

To obtain an accurate energy consumption value, it is strongly recommended to directly measure the electricity consumed during the extraction process using a clamp ammeter. The nominal power rating reported on the instrument should not be used, as it refers to maximum absorption parameters and does not reflect actual operational consumption.

Duration:

Simply insert the relative duration of the instrument use inside the process expressed in minutes.

4.5 Waste



Waste ☑

Product kg ⓘ
Enter product amount

Matrix kg
Enter matrix amount

Solvent kg ⓘ
Enter solvent amount

Water kg ⓘ
Enter water amount

Additive kg ⓘ
Enter additive amount

Figure 12. Waste compilation window.

In this EF, the User is required to enter the data necessary to calculate the Process Mass Intensity (PMI) of the extraction process (**Figure 12**). The required quantity data will be then elaborated according to the following equation:

$$PMI = \frac{Biomass\ (kg) + Solvent\ (kg) + Water\ (kg) + Additive\ (kg)}{Product\ (kg)}$$

The PMI expresses the total mass of materials consumed per unit mass of product obtained.

Total solvent and additive masses must first be included in the global PMI calculation. In the subsequent compilation step, the User will be asked to specify the relative quantity of each solvent and additive contributing to the final waste, normalized with respect to the processed biomass (**Figure 13**). The list of solvents and additives appearing in this section is automatically generated based on the entries previously provided in the “Solvents & Additives” section.

Any mass that is recycled during the process, whether originating from a solvent or an additive, may be excluded from the calculation. Only the net consumed quantities must be considered. For example, when using a rotary evaporator to recover a solvent, the fractionated and reusable solvent, if properly recovered and reintroduced into the process, should be excluded from the waste calculation. However, the cooling water required for the condenser must be included, as it represents an operational input

contributing to the overall mass intensity. Indeed, water is accounted for separately from solvents to include auxiliary water used for heating or cooling during the process.

Solvent List:

Ethanol

free ☒ present

Amount (S/L)

1:

Additive List

Citric Acid

free ☒ present

Amount (% on matrix weight)

Figure 13. Discarded specific quantity for each chemical used in the process.

4.6 Product

The final EF to complete relates to the characteristics of the extracted product (Figure 14). In this section, the User must report the selectivity of the product, whether it is a dry extract or another form (e.g., solubilized in deep eutectic solvent). Selectivity should be expressed as milligrams of target compound per gram of extract (mg target compound / g extract).

In addition, the User is required to provide a list of possible contaminants. These are substances that may originate from the solvents or additives previously entered in the “Solvents & Additives” section. If traces of these chemicals are present in the extract, the User simply marks their presence. Unlike the Waste section, where quantities are used in calculations, here only the presence or absence of contaminants is recorded. The corresponding hazard points for each detected contaminant are automatically applied as penalty points in the final evaluation.

Product



Selectivity mg Target Compound/ g Extract

Enter Selectivity

Contaminant List:

Ethanol

free present

Citric Acid

free present

Figure 14. Product compilation window

5. Ready to submit

After completing all required fields, click *Submit* to finalize your entry. Before submission, the system will prompt you to accept the Privacy Policy, which can be viewed by clicking on *Privacy Policy* (**Figure 15**). If any mandatory fields are missing, the software will prevent submission and highlight the incomplete entries in red within the form.

☒ I read the privacy policy and accept the terms

Submit 

Figure 15. Privacy Policy disclaimer to accept prior submission.

Privacy Policy

Pursuant to Regulation (EU) 2016/679 (GDPR), we inform you that the data provided through this form will be used to create a public database on organic extraction techniques, freely accessible on this website in the Database section. The purpose of this initiative is to develop a comparative metric on the environmental impact of these procedures and to support scientific analysis.

The designated Data Controller is Giorgio Grillo. All data are processed securely and stored on Firebase servers.

No personal identifiers are intentionally collected. However, certain data fields, such as DOI or institutional affiliation, may allow indirect identification of contributors.

Participants may exercise their rights under Articles 15 to 22 of the GDPR, including the rights of access, rectification, erasure, restriction of processing, and objection, by contacting six.score.metric@gmail.com.

This website uses only technical cookies required for its proper functioning and does not employ profiling or tracking cookies.

This privacy notice may be updated periodically to reflect regulatory or procedural changes. Participants are encouraged to review it regularly to remain informed about data processing practices.

6. Summary Overview

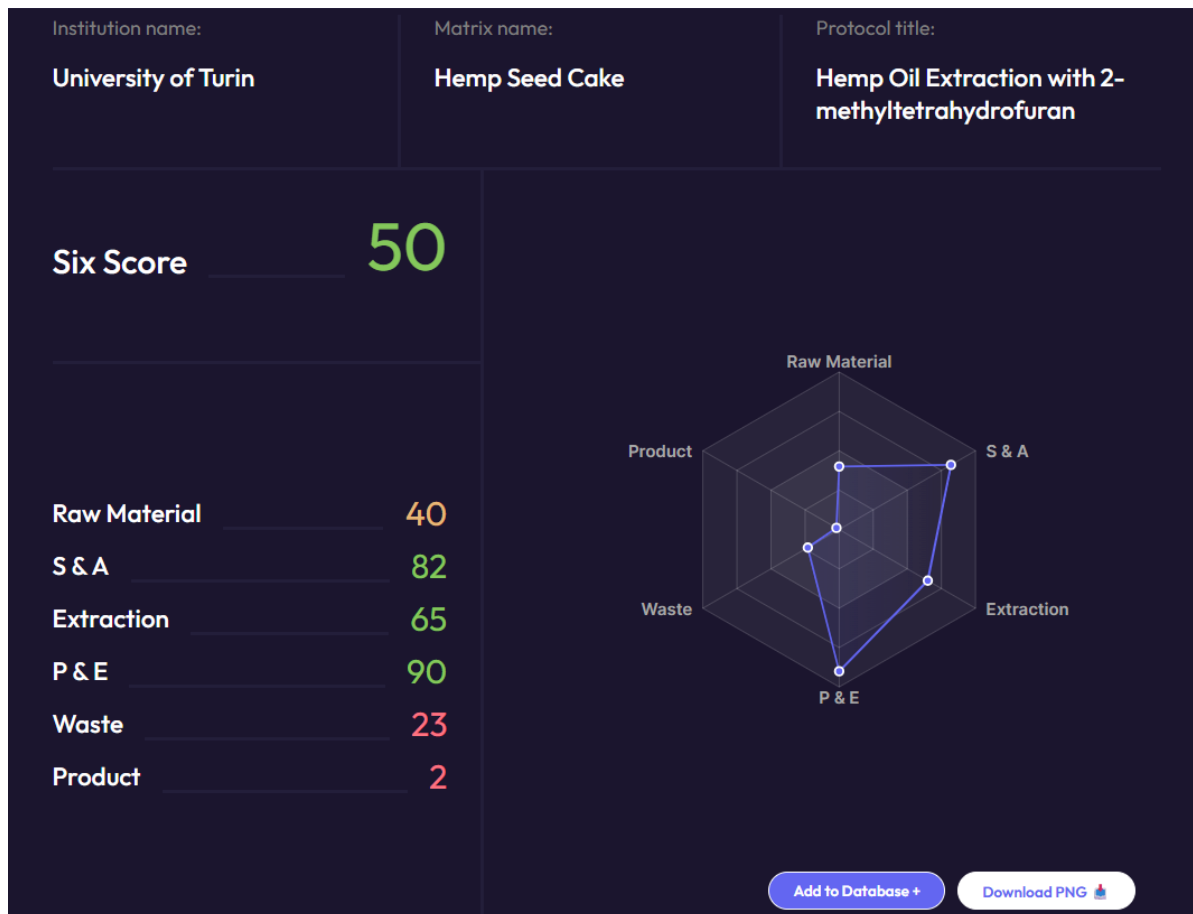


Figure 16. Visual outcome of the sustainability assessment.

Once the form is successfully submitted, the summary screen will appear (**Figure 16**). The visual outcome will display the following information:

- Name of the User Institution together with the Matrix and Protocol evaluated;
- Total SIX Score and scores for each EF, presented in tabular format and visualized in a radar chart;
- Two Options “Add to Database” and “Download PNG”;

If you choose to add the results to the database, this is only possible if the DOI of the related publication has been provided. Once shared, Users can re-open the project to view the outcomes, or alternatively, copy the data to start a new survey based on the registered values. It is important to note

that once a survey is added to the database, it cannot be modified. However, Users may request deletion at any time by contacting the SIX Score support email.

Scrolling further down, a detailed breakdown of the results is provided, showing the penalty points assigned to each subcategory (**Figure 17**). This feature enables the User to identify strengths and weaknesses in the process, facilitating targeted improvements.

Extraction	65
Depletion Efficiency	(71.86) 6.8
Solvent Impact	(2.00) 5.0
Carbon Economy	(0.12) 20.8
Duration	(70.00) 2.5

Figure 17. Definition of subcategory for the elemental factor compilation window.

At the bottom of the page, additional options are available (**Figure 18**). By clicking “*Modify Current*”, previously entered information are loaded, bringing back to the evaluation form enabling updates as needed. If the user wants to load other survey previously compiled and downloaded as json format can click on “Load Data to Current” and the software will show you the related results window for the selected process.

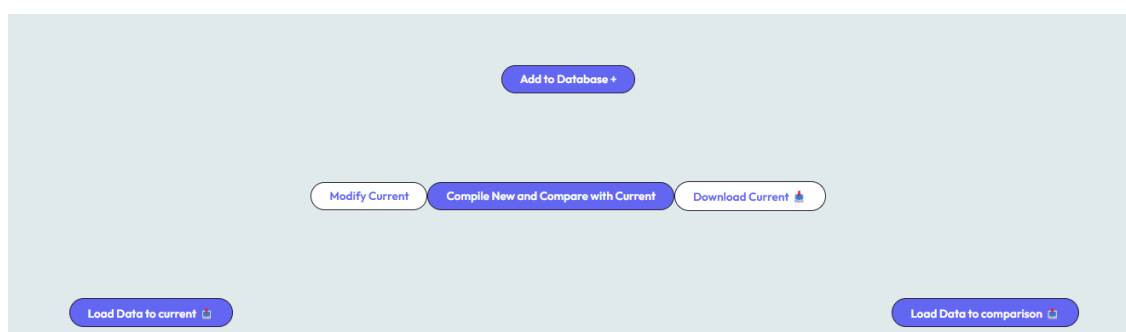


Figure 18. Further actions following the sustainability assessment.

6.1 SIX Score Comparison

To facilitate the comparison of two sustainability assessments, the “*Compile New and Compare*” feature allows the user to create a new survey for a process and directly compare it with the current one. Alternatively, users can “*Load Data to Current Comparison*” to compare a previously downloaded survey (in. json format) with the current results. As an output the radar graph will

compare the two processes for each EF defining “Current Protocol” as the first one loaded and “*Comparison*” for the second loaded (**Figure 19**). Following information such as protocol name and matrix are reported priorly the description of each EF which are compared in two distinct columns (**Figure 20**). The function “*Comparison*” allows the user to compare only two process per time.

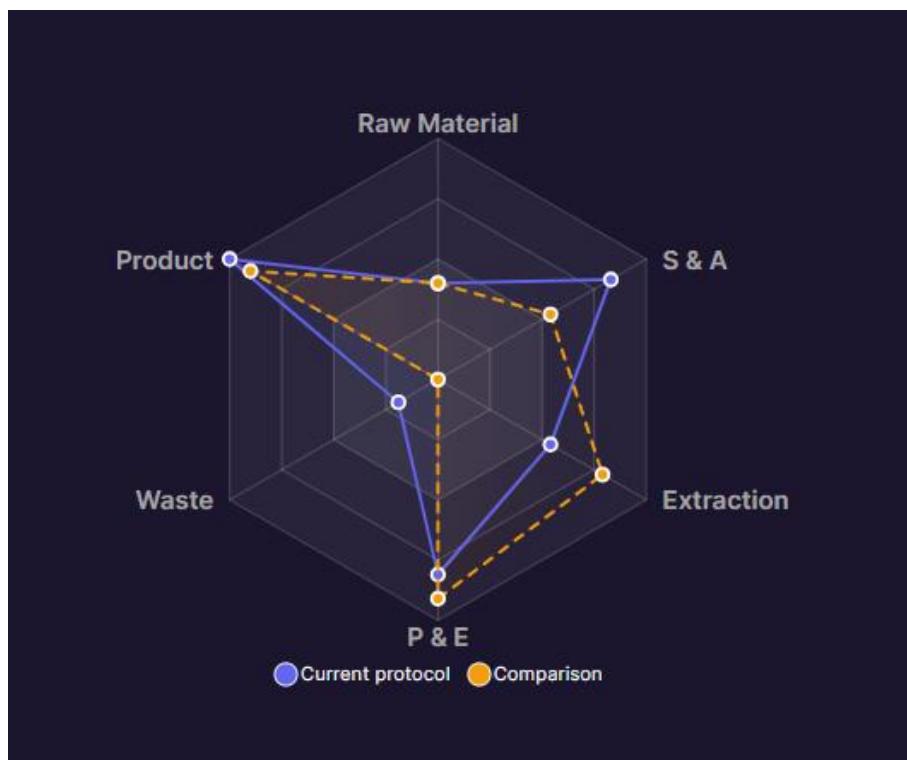


Figure 19. Comparison radar graph for two sustainability assessments.

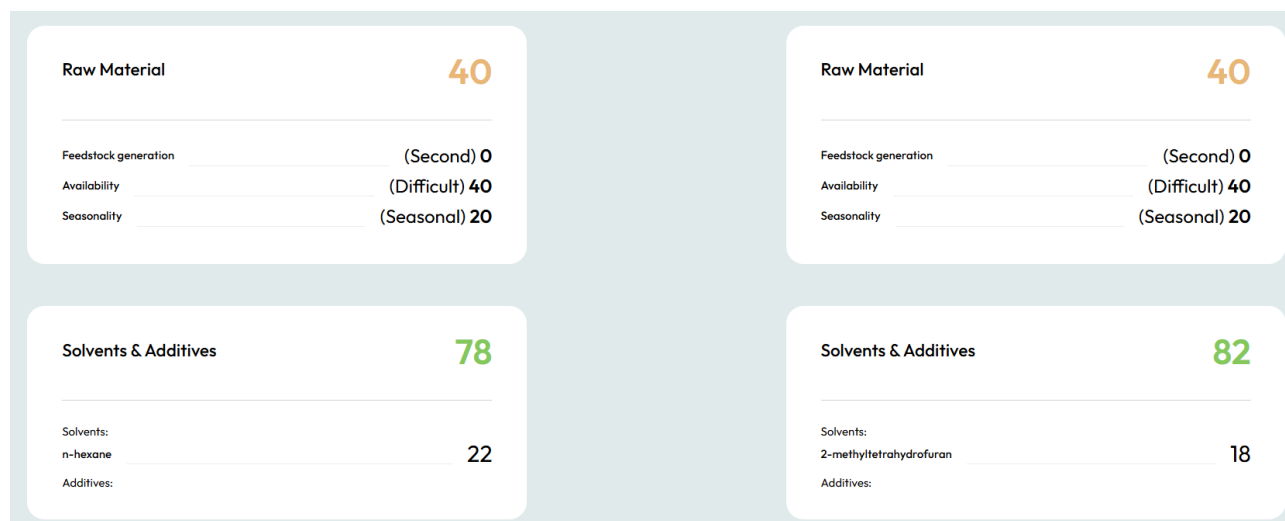


Figure 20. Comparison for different EF together with subcategories for two sustainability assessments.

7. Database

The SIX Score Hub database is a structured, traceable, community-driven repository of SIX Score assessments. It enables aggregation of population-level trends for benchmarking and periodic refinement of thresholds and scoring functions and provides a robust foundation for future workflows while preserving full data origin.

Users can access the database from the selection panel in the top-right section of the website. Each sustainability assessment included in the database is registered only after user approval and upon submission of a valid DOI together with the assessment, ensuring that only peer-reviewed protocols are included (**Figure 21**).

By clicking on “*See Report*”, the website directs the user to the results page of the selected sustainability assessment. From this page, users can review the overall scores as well as those related to the different subcategories. If additional details are needed, users can access the underlying data and the author’s notes by selecting “*Modify Current*”.

Although this option opens the “*Modify Current*” section, the user cannot overwrite the stored data; the feature is intended solely to allow deeper exploration and understanding of the assessed process.

Institution Name	Matrix Name	Protocol Title	DOI	Raw Material	S & A	Extraction	P & E	Waste	Product	SixScore	Detail
Department of Drug Science and Technology, UniTO	Pomegranate Peels	MAE + NF	10.3390/antiox12101796	40	100	89	71	87	91	80	See Report
Department of Drug Science and Technology, UniTO	Blueberry Pomace	NADES (CHCl ₃ :LA)	10.3390/antiox9111069	80	92	80	90	100	0	74	See Report
Department of Drug Science and Technology, UniTO	Pomegranate	MAE (water)	10.3390/antiox12101796	40	100	89	71	46	80	71	See Report
Department of Drug Science and Technology, UniTO	Pomegranate Peels	Conventional (EtOH)	10.3390/antiox12101796	40	79	91	66	29	90	66	See Report
Department of Drug Science and Technology, UniTO	Blueberry Pomace	MASWE	10.3390/ijms252011032	80	100	66	81	2	59	65	See Report
Department of Drug Science and Technology, UniTO	Blueberry Pomace	MAE	10.3390/ijms252011032	80	94	72	81	6	52	64	See Report
Department of Drug Science and Technology, UniTO	Blueberry Pomace	Conventional (EtOH+HCl)	10.3390/ijms252011032	80	77	73	75	4	73	64	See Report
Department of Drug Science and Technology, UniTO	Blueberry Pomace	MAE+MASWE	10.3390/ijms252011032	80	94	86	62	4	52	63	See Report

Figure 21. Database presentation list.